

A volume-normalized linear contraction obtained by a limit

MathlibAnnex.PluckerRecovery.nonempty_limitCertificate_of_pluckerBodies_eq

theorem

Takes a compact operator limit preserving the exact volume equation.

Statement

Let $n = m + 1 > 0$ and $V = \mathbb{R}^n$ with the reference sup norm and standard Lebesgue measure λ . The continuous norms p, q have fixed positive comparison constants

$$a_p \|x\|_\infty \leq p(x) \leq b_p \|x\|_\infty, \quad a_q \|x\|_\infty \leq q(x) \leq b_q \|x\|_\infty.$$

Set $K_p = \{x : p(x) \leq 1\}$, $K_q = \{x : q(x) \leq 1\}$ and $v_p = \lambda(K_p)$, $v_q = \lambda(K_q)$, as positive finite real numbers. For a real linear map $A : V \rightarrow \mathbb{R}^N$, let $A_{ij} = (Ae_j)_i$ be its matrix in the standard bases. The minor-coordinate space has one coordinate for each increasing n -tuple I of distinct rows from $\{1, \dots, N\}$, and

$$\Pi_n(A)_I = \det(A_{I,*}).$$

Define the signed generator set and its real convex hull by

$$\mathcal{G}_p^N = \left\{ \sigma v_p \Pi_n(A) : \begin{array}{l} A : V \rightarrow \mathbb{R}^N \text{ is real linear, } \sigma \in \{1, -1\}, \\ \|Ax\|_\infty \leq p(x) \quad (x \in V) \end{array} \right\},$$

$$\mathcal{P}_p^N = \text{conv}_{\mathbb{R}} \mathcal{G}_p^N.$$

Define $\mathcal{G}_q^N, \mathcal{P}_q^N$ in the same way using q, v_q . The generators are vectors of minors, not linear maps. Assume $\mathcal{P}_p^N = \mathcal{P}_q^N$ for every N . There is one real linear $L : V \rightarrow V$ with

$$\forall x, q(Lx) \leq p(x), \quad L(K_p) \subseteq K_q, \quad v_p |\det L| = v_q,$$

$$\det L \neq 0, \quad L \text{ injective and surjective.}$$

Assumptions

The positive-dimensional coordinates, norm models and equality of every finite body are the inputs. No distortion parameter is supplied.

Conclusion

The same real linear bijection L is contractive from the norm p to the norm q and has the exact determinant-volume normalization. Only ball inclusion is asserted here; equality of the two ball images is proved separately.

Proof route

Choose vanishing distortions, bound the maps in one compact operator ball, and pass the estimate and determinant equation to the same subsequence limit.

▼ Proof steps

1. **Choose and bound the sequence.** Set $\varepsilon_k = 1/(k+2)$. For each k , linear recovery at the positive parameter ε_k gives

$$(1 - \varepsilon_k)q(L_k x) \leq p(x), \quad v_p |\det L_k| = v_q.$$

Because $0 < \varepsilon_k \leq 1/2$ and $q \geq 0$,

$$\frac{1}{2}q(L_k x) \leq (1 - \varepsilon_k)q(L_k x) \leq p(x),$$

$$a_q \|L_k x\|_\infty \leq q(L_k x) \leq 2p(x) \leq 2b_p \|x\|_\infty.$$

Consequently $\|L_k x\|_\infty \leq (2b_p/a_q)\|x\|_\infty$. These are the half-distortion estimate and the reference-norm estimate, applied to the selected certificates.

2. **Apply compactness in the operator space.** For real linear maps $T : V \rightarrow V$, use the reference operator norm $\|T\|_{\text{op}} = \sup_{\|x\|_\infty \leq 1} \|Tx\|_\infty$. All maps belong to the same closed operator ball $\{T : \|T\|_{\text{op}} \leq 2b_p/a_q\}$. The real space of linear maps $V \rightarrow V$ is finite-dimensional, so this ball is compact. The compact subsequence construction returns strictly increasing k_j and one L with $L_{k_j} \rightarrow L$ in operator norm. For each fixed x ,

$$\|L_{k_j} x - Lx\|_\infty \leq \|L_{k_j} - L\|_{\text{op}} \|x\|_\infty \rightarrow 0.$$

Strict increase gives $k_j \rightarrow \infty$ and hence $\varepsilon_{k_j} \rightarrow 0$. In compact subsequence selection, the input sequence lies in one compact set in a first-countable space; the output is a limit in that set and a strictly increasing subsequence. The continuous evaluation map and its RHS is $\text{ev}_x(T) = Tx$, giving the same pointwise limit.

3. **Pass the norm estimate to that limit.** Continuity of q gives $q(L_{k_j} x) \rightarrow q(Lx)$. Therefore

$$(1 - \varepsilon_{k_j})q(L_{k_j} x) \rightarrow q(Lx).$$

Every term is at most $p(x)$, so $q(Lx) \leq p(x)$. If $x \in K_p$, then $q(Lx) \leq 1$, giving $L(K_p) \subseteq K_q$. These are the limit contraction and its ball inclusion.

4. **Pass the volume identity to the same map.** The determinant is continuous in the entries, so $\det L_{k_j} \rightarrow \det L$. Thus

$$v_p |\det L| = \lim_{j \rightarrow \infty} v_p |\det L_{k_j}| = v_q.$$

This is the limit determinant-volume equation. Since $v_q > 0$, the determinant cannot vanish. It makes L injective, and an injection from finite-dimensional V to itself is surjective. Store this same L in every field.

The subsequence and its convergence are construction evidence, not extra public fields. Operator compactness uses the reference sup norm; contraction compares the stored norms p, q .

Main citations

- Exact declaration and source
- linear recovery
- the half-distortion estimate
- the reference-norm estimate
- compact subsequence construction
- compact subsequence selection
- continuous evaluation map and its RHS
- the limit contraction
- its ball inclusion
- the limit determinant-volume equation
- The seven-field limit certificate
- the signed generator set
- its real convex hull

Lean source signature (exact)

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-- Equality of all finite Plücker bodies yields a limit recovery certificate. -/
theorem nonempty_limitCertificate_of_pluckerBodies_eq {m : ℕ}
  (MX MY : EquivalentSeminorm (Fin (m + 1) → ℝ))
  (hBodies : ∀ N : ℕ, PluckerBody.body MX N = PluckerBody.body MY N) :
  Nonempty (LimitCertificate MX MY)
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In the source	Mathematical meaning
<code>Fin (m + 1) → ℝ; →L[ℝ]</code>	The reference $V = \mathbb{R}^n, n = m + 1 > 0$, with sup norm; arrows mean continuous real linear maps. Source indices $0, \dots, n - 1$ correspond to formula indices $1, \dots, n$.
<code>MX MY : EquivalentSeminorm (Fin (m + 1) → ℝ); MX.p; MY.p</code>	The two continuous norms p, q and their positive comparison constants. The applications <code>MX.p x</code> and <code>MY.p x</code> mean $p(x), q(x)$.
<code>MX.closedUnitBall; MY.closedUnitBall; MX.closedUnitBallVolume; MY.closedUnitBallVolume</code>	The sets K_p, K_q and their real Lebesgue volumes v_p, v_q . Real volumes are the <code>toReal</code> of the finite extended measures.
<code>hBodies : ∀ N : ℕ, PluckerBody.body MX N = PluckerBody.body MY N</code>	For every nonnegative N , the real convex hulls $\mathcal{P}_p^N$ and $\mathcal{P}_q^N$ are equal. The source body is the convex hull of the vectors $\pm v_p \Pi_n(A)$, where $A : V \rightarrow \mathbb{R}^N$ is real linear and $\ Ax\ _\infty \leq p(x)$ for every x . The target body uses q, v_q in place of p, v_p .
<code>Nonempty (LimitCertificate MX MY)</code>	Existence of one record with all seven fields below, without a supplied ε .

There is no supplied error parameter in this theorem. From the two norm models and hBodies, the final Nonempty (...) asserts existence of one map with all seven properties below.

Related structure, shown separately: The seven-field limit certificate.

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structure LimitCertificate {m : ℕ}
  (MX MY : EquivalentSeminorm (Fin (m + 1) → ℝ)) where
  linearMap : (Fin (m + 1) → ℝ) →L[ℝ] (Fin (m + 1) → ℝ)
  seminorm_linearMap_le : ∀ x, MY.p (linearMap x) ≤ MX.p x
  image_closedUnitBall_subset : linearMap '' MX.closedUnitBall ⊆ MY.closedUnitBall
  closedUnitBallVolume_mul_abs_det : MX.closedUnitBallVolume * |ContinuousLinearMap.det
linearMap| = MY.closedUnitBallVolume
  det_ne_zero : ContinuousLinearMap.det linearMap ≠ 0
  injective : Function.Injective linearMap
  surjective : Function.Surjective linearMap

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In the source	Mathematical meaning
linearMap	The chosen continuous real linear map $L : V \rightarrow V$.
seminorm_linearMap_le	$q(Lx) \leq p(x)$ for every $x \in V$.
image_closedUnitBall_subset	$L(K_p) \subseteq K_q$: every vector Lx with $p(x) \leq 1$ satisfies $q(Lx) \leq 1$.
closedUnitBallVolume_mul_abs_det	$v_p \det L = v_q$, for this same L and these finite real volumes.
det_ne_zero	$\det L \neq 0$.
In the source	Mathematical meaning
injective	If $Lx = Ly$, then $x = y$.
surjective	Every $y \in V$ is Lx for some $x \in V$.

Exact definitions: the signed generator set; its real convex hull.

Surrounding assumptions and aliases.

▼ Exact content identity

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